

A* Examination Guide

Built from 19 past papers and their mark schemes, January 2020 to June 2026

This guide is built from the P3 question papers and mark schemes themselves. Methods are set out the way the mark schemes award M and A marks, and every warning comes from a note that examiners applied: “A0”, “withhold”, “do not allow”, “answers with no working score no marks”. Use the formula booklet for what it gives you, and learn the rest.

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Papers analysed: Jan 2020, Oct 2020, Jan 2021, Jun 2021, Oct 2021, Jan 2022, Jun 2022, Oct 2022, Jan 2023, Jun 2023, Oct 2023, Jan 2024, Jun 2024, Oct 2024, Jan 2025, Jun 2025, Oct 2025, Jan 2026, Jun 2026. Papers are cited by series, e.g. (Jun 24).

1 HOW P3 IS BUILT

Format: 1 hour 30 minutes, 75 marks, 9–10 questions: about 1.2 minutes per mark. Calculators are allowed but cannot do symbolic algebra, and many questions now say “Solutions relying on calculator technology are not acceptable”. When they do, an answer with no working scores 0, even if it is right. Each paper mixes short skill questions (functions, logs, sign change) with three or four long, multi-part questions worth 9–13 marks.

Topic	Papers (of 19)	What is asked most
Functions: composite, inverse, range	19	$fg(a)$, f^{-1} , range and domain, solve $f(x) = f^{-1}(x)$, $gf(a) = k$
Trigonometric identities and equations	19	prove an identity, then “hence solve” in a given range
Differentiation	19	chain, product, quotient; stationary points; tangent/normal; dx/dy
Exponential models in context	18	initial value, limit, solve for t , rate of change, find constants
Logarithmic graphs (log y against x or log x)	18	line equation $\rightarrow y = ab^x$ or $y = ax^n$; interpret constants
Integration	18	standard forms, f'/f , trig identities, algebraic division, exact areas
Modulus graphs and transformations	18	vertex, solve equation/inequality, roots for $f(x) = k$, $af(x + b)$
Numerical methods	17	sign change, rearrange to $x = g(x)$, iterate, verify a root to 3 d.p.
$x = f(y)$: find dy/dx in terms of x	12	differentiate w.r.t. y , invert, use a Pythagorean identity
$R \cos(x \pm \alpha)$ / $R \sin(x \pm \alpha)$	11	exact R , α in radians, max/min of a related function

The recurring long questions

- **Exponential model (8–13 marks):** every paper except Jan 21 and Oct 20 has one, from toads (Jan 20) to car prices (Jun 26). The parts follow the same pattern: value at $t = 0$, limiting value, solve for t , differentiate, interpret.
- **Prove a trig identity, then “hence solve” (6–9):** Jan 21, Jun 23, Oct 23, Jan 24, Jun 25, Oct 25, Jun 26.
- **Iteration with sign change (6–9):** stationary point $\rightarrow x = g(x) \rightarrow x_2, x_n \rightarrow$ interval check.
- **Modulus graph with a line (7–10):** vertex, inequality, and the range of k or m for a given number of intersections.
- **$x = f(y)$ curves (5–9):** Jan 21, Jun 21, Oct 21, Jan 22, Oct 23, Jun 24, Jan 25, Jun 25, Jan 26, Jun 26.

2 TWENTY MARK-SCHEME RULES FOR P3

1. **“Solutions relying on calculator technology are not acceptable”:** show the algebra. Correct answers with no working score no marks (Jun 22, Oct 23, Jun 24, Jan 25, Jun 25).
2. **“Hence”** means use the previous part. “This is a ‘hence’ question, so starting from scratch scores B0 M0 A0” (Jan 26).
3. **Radians or degrees:** α is in radians when asked ($\alpha = \text{awrt } 22.6^\circ$ is A0). Answers in radians to a degrees question score A0. Sign-change values computed in degrees mode score A0 (Jan 25).
4. **R is exact:** $R = \sqrt{17}$, not 4.12 (“do not allow decimals for this mark”). $R = -25$ must be rejected.
5. **Sign change needs four things:** both values, “change of sign”, “continuous”, and a conclusion. “The root is continuous” or “ x is continuous” loses the A mark.
6. **Iteration answers are to the stated d.p.** and keep trailing zeros: $x_5 = 0.8110$; “0.811 is A0” (Jan 20). $x_6 = 2.155$ was cao (Jun 21).
7. **Inverse functions must be in terms of x .** $f^{-1}(x) = \dots$ is fine; “ $y = \dots$ ” alone is not accepted.
8. **Ranges use $f(x)$ or y , never x :** $0 \leq f(x) < 2/3$. Writing the range in x scores A0 (Oct 22).
9. **Do not throw away valid roots:** rejecting $x = 0$ in $f^{-1}(x) = f(x)$ scored A0 (Jun 25). Extra unrejected solutions in range also lose the final A mark.
10. **Modulus in ln:** $\int 1/(x - 5) dx = \ln|x - 5|$. Evaluating ln of a negative number without modulus signs loses the final A1 (Jun 21).
11. **Indefinite integrals need + c and simplest form;** a definite answer must not still carry $\int, dx$ or $+ c$ (Jun 24).
12. **“Show that” answers:** every step, no errors, no circular reasoning. In trig proofs that need it, a Pythagorean identity must be used (Oct 23).
13. **Rates in models:** differentiate first, then substitute. Substituting t before differentiating loses the mark (Jan 26).
14. **Interpreting constants:** $b = 1.19$ means “increases by 19% each year”; p = the initial value or the value when $r = 1$. “Rate of change” or “the gradient” is rejected (Oct 23).
15. **$\log_{10}$ is not ln:** “do not accept $\ln N$ for $\log_{10} N$ ” (Jun 22). Write the line in the given variables: $\log_{10} N = 3.08 + 0.154t$.
16. **Context answers need units and sense:** “a loss of £830 000”; a negative value was A0 (Oct 22). Give times in the unit asked (years and months).
17. **Quotient rule the right way round:** $(vu' - uv')/v^2$. Reversed numerators lose the marks (Jun 24).
18. **Equations of lines in the requested form:** $ax + by + c = 0$ with integers and “ $= 0$ ” seen (Jun 22).
19. **Accuracy:** 3 s.f. unless told otherwise. When a decimal is asked for, an exact answer only is A0 (Jan 26); when “exact” is asked for, a decimal is A0.
20. **Notation:** brackets and powers matter. $\sin x^2$ is not $\sin^2 x$; repeated notational errors cost the final A mark (Jan 22, Oct 24).

3 ALGEBRAIC FRACTIONS AND DIVISION

- Factorise every numerator and denominator first, then cancel; find a common denominator to add or subtract. Typical target: “show that $f(x) = 7x/(x + 3)$ ” (Jun 25) or “ $(ax + b)/(cx + d)$ ” (Jan 21, Jan 25).
- Improper fraction $\rightarrow$ polynomial + proper fraction before integrating or differentiating. Divide, or write the identity and compare coefficients.

WORKED EXAMPLE · DIVISION BEFORE INTEGRATING (JUN 24)

$g(x) = (2x^2 + 5x - 8)/(x + 2)$. Since $(x + 2)(2x + 1) = 2x^2 + 5x + 2$, $g(x) = 2x + 1 - 10/(x + 2)$. Then $\int g(x) dx = x^2 + x - 10 \ln|x + 2| + c$, and the limits give an answer of the form $\alpha + \beta \ln 3$.

For a quadratic divisor (Jan 23, Jan 25, Jan 26) write $f(x) \equiv Ax^2 + Bx + C + (Dx + E)/(\text{divisor})$, multiply through, and compare coefficients or substitute values. Misreading the divisor scores nothing.

4 FUNCTIONS

DEFINITIONS AND FACTS USED EVERY PAPER

$fg(x)$ means apply g first, then f . The domain of fg is the part of the domain of g whose outputs lie in the domain of f .

f^{-1} exists only if f is one-to-one. Domain of $f^{-1} = \text{range of } f$; range of $f^{-1} = \text{domain of } f$. The graph of $y = f^{-1}(x)$ is the reflection of $y = f(x)$ in $y = x$.

To find f^{-1} : write $y = f(x)$, make x the subject (collect x terms, factorise), then swap to $f^{-1}(x) = \dots$ in terms of x .

Increasing/decreasing: show $f'(x) > 0$ (or < 0) for all x in the domain and state the conclusion about f (Jan 22, Jan 24, Jun 24).

WORKED EXAMPLE · JUN 25 Q1 (8 MARKS)

$f(x) = 2x/(3x + 1)$, $x \geq 0$; $g(x) = 4 + x^2$.

(a) $gf(1)$: $f(1) = 1/2$, so $g(1/2) = 17/4$. (b) $f(0) = 0$ and $f(x) \rightarrow 2/3$ as $x \rightarrow \infty$, so $0 \leq f(x) < 2/3$.

(c) $y(3x + 1) = 2x \Rightarrow x(3y - 2) = -y \Rightarrow f^{-1}(x) = x/(2 - 3x)$.

(d) $f^{-1}(x) = f(x) \Leftrightarrow f(x) = x$ here: $2x = x(3x + 1) \Rightarrow x = 0$ or $x = 1/3$. Both are needed; rejecting $x = 0$ scored A0.

WORKED EXAMPLE · JAN 22 Q6

$f(x) = (5x - 3)/(x - 4)$, $x > 4$. $f'(x) = [5(x - 4) - (5x - 3)]/(x - 4)^2 = -17/(x - 4)^2 < 0$ for all $x > 4$, so f is decreasing.

$ff(x) = (22x - 3)/(x + 13)$. As $x \rightarrow 4^+$, $ff \rightarrow 5$; as $x \rightarrow \infty$, $ff \rightarrow 22$. Range: $5 < ff(x) < 22$.

- Solving $gf(a) = k$ or $fg(x) = k$: either substitute and solve, or apply inverses in the right order: $f(a) = g^{-1}(k)$. “Undoing g once” when g appears twice scored nothing (Jan 25).
- Ranges of trig and exponential compositions come from the range of the inner function, e.g. $4 \sin(\dots) \in [-4, 4]$ (Oct 22).
- Where $y = f(x)$ meets $y = f^{-1}(x)$ for an increasing f , solve $f(x) = x$ (Jun 23).

5 MODULUS GRAPHS AND TRANSFORMATIONS

Transformation	Effect on a point (p, q)
$y = f(x + b)$	$(p - b, q)$
$y = af(x)$	(p, aq)
$y = f(ax)$	$(p/a, q)$

Transformation	Effect on a point (p, q)
$y = af(x + b)$ (vertex questions: Oct 20, Oct 21, Jan 26)	$(p - b, aq)$
$y = f(x) $	parts below the x-axis are reflected above it
$y = f(x)$	the part for $x \geq 0$ is reflected in the y-axis

METHOD · SOLVING MODULUS EQUATIONS AND INEQUALITIES

Sketch both graphs first. For $|ax + b| = \text{line}$, solve $ax + b = \text{line}$ and $-(ax + b) = \text{line}$ separately, then check each solution against the sketch: a solution that does not satisfy its own case must be rejected.

Inequality: find the critical values the same way, then choose the region from the sketch (inside for $<$, outside for $>$).

WORKED EXAMPLE · JUN 24 Q1

$f(x) = 2|x - 5| + 10$. Vertex $P(5, 10)$. Under $y = 3f(x - 2)$, $P \rightarrow (7, 30)$.

Solve $2|x - 5| + 10 > 6x$. For $x \geq 5$: $2x - 10 + 10 > 6x$ gives $x < 0$, impossible. For $x < 5$: $-2x + 10 + 10 > 6x$ gives $x < 2.5$. Solution: $x < 2.5$.

- Number of roots of $f(x) = k$: read from the sketch. For a V-shaped graph with vertex (p, q) and a restricted domain, two roots need k strictly between q and the endpoint value.
- A line $y = mx$ through the origin meeting a modulus graph once or twice (Jan 26, Jun 25, Oct 25): compare the line's gradient with the gradients of the arms, and use the vertex as the critical case.
- Sketches: show the vertex and all axis intercepts as coordinates; asymptotes as equations.

6 EXPONENTIALS, LOGARITHMS AND MODELLING

LAWS TO USE WITHOUT THINKING

$\ln$ and e^x are inverses: $e^{\ln a} = a$, $\ln e^a = a$. $\ln a + \ln b = \ln ab$; $\ln a - \ln b = \ln(a/b)$; $k \ln a = \ln a^k$. $d/dx(e^{kx}) = ke^{kx}$; $d/dx(a^x) = a^x \ln a$.

To solve $Ae^{kt} = B$: $e^{kt} = B/A \Rightarrow t = (1/k) \ln(B/A)$. To make k the subject use $\ln$, not $\log_{10}$ (Jan 21).

Exponential models in context

Part	What to do	Past-paper example
value at the start	put $t = 0$ ($e^0 = 1$)	$N = 900e^{0.12t}/(2e^{0.12t} + 1)$: $N(0) = 300$ (Jan 20)
limit as $t \rightarrow \infty$	divide through by e^{kt} , or note $e^{-kt} \rightarrow 0$	$N \rightarrow 900/2 = 450$, so it can never reach 500
solve for t	clear the fraction, isolate e^{kt} , take $\ln$	$N = 420$: $60e^{0.12t} = 420 \Rightarrow t = \ln 7/0.12 = 16.22$
find a constant	use the two given conditions	$F = 350e^{kt}/(9 + e^{kt})$, $F(15) = 200 \Rightarrow e^{15k} = 12$, $k = (1/15) \ln 12$ (Jan 23)
rate of change	differentiate, then substitute	$N = 600e^{0.3t}/(2 + e^{0.3t})$: $dN/dt = 360e^{0.3t}/(2 + e^{0.3t})^2$ (Jun 21)
time in context	convert decimals of a year	$t = \ln 10/0.3 = 7.675$ years = 7 years 8 months (Jun 21)

WORKED EXAMPLE · JAN 21 Q5 (OVEN)

$\theta = A - 180e^{-kt}$. $\theta(0) = 18 \Rightarrow A = 198$. $\theta(5) = 90 \Rightarrow e^{-5k} = 108/180 = 0.6 \Rightarrow k = (1/5) \ln(5/3) = 0.1022$.

$\theta(9) = 198 - 180e^{-9k} = 126$ °C (3 s.f.). Rate: $d\theta/dt = 180k e^{-kt} = 7.33$ °C min^{-1} at $t = 9$.

Same structure: processor $\theta = 21 + Ae^{-kt}$ (Jun 25): $A = 54$, $k = 0.521$, and $d\theta/dt = -9$ gives $T = 2.19$.

- “Explain why the model is not suitable” (Oct 23): the model’s limit (125 thousand) is below the expected 150 thousand. Use the numbers.
- “Explain why the temperature cannot fall below 10 °C” (Jan 24): $Ae^{-Bt} > 0$ for all t , so $T > 10$.
- dH/dt in terms of H (Jan 25): $H = 280e^{-0.05t} + 24 \Rightarrow dH/dt = -0.05(H - 24) = 1.2 - 0.05H$.

Logarithmic graphs

Graph plotted	Model	Gradient and intercept give
log y against x (or t)	$y = ab^x$	log y = log a + x log b: intercept = log a, gradient = log b
log y against log x	$y = ax^n$	log y = log a + n log x: intercept = log a, gradient = n

WORKED EXAMPLES

Jun 21: line through (0, 0.32) and (8, 0.56): $\log_{10} A = 0.32 + 0.03t$, so $p = 10^{0.32} = 2.089$ and $q = 10^{0.03} = 1.072$. Rate at $t = 6$: $dA/dt = pq^t \ln q = 0.22 \text{ m}^2/\text{day}$.

Jun 25: line through (-3.5, 0) and (0, -2): $\log_{10} y = -(4/7) \log_{10} x - 2$, so $y = 0.01x^{-4/7}$.

Oct 21 (tree): $\log_{10} M = 1.93 \log_{10} r + 0.684 \Rightarrow M = 4.83r^{1.93}$. For $r = 45$, $M = 7500 \text{ kg}$ (2 s.f.).

Jan 22: $\log_{10} x = 2.74 - 0.079t \Rightarrow x = 550 \times 1.2^{-t}$. For $x = 400 \times 1.4^{-t}$, $dx/dt = -400 \ln 1.4 \times 1.4^{-t} = -25$ at $t = 5$.

INTERPRETING THE CONSTANTS · OCT 21, JAN 22, OCT 23, JAN 26, JUN 26

a or p ($y = ab^x$): the initial value, in context: “the amount of antibiotic in the bloodstream when the dose was given”. For $y = px^q$, p is the value of y when $x = 1$, e.g. “the mass of a tree of base radius 1 cm”.

b or q: the growth factor per unit time, stated as a percentage with context: “the number of households recycling increases by 19% each year”; “the area of reef is multiplied by 0.986 each year (falls by 1.4%)”. “Rate of change” and “the gradient” are rejected.

7 TRIGONOMETRY I: RECIPROCAL FUNCTIONS AND IDENTITIES

Function	Definition	Key facts
sec x	$1/\cos x$	$ \sec x \geq 1$; undefined where $\cos x = 0$
cosec x	$1/\sin x$	$ \csc x \geq 1$, so cosec $x = -2/3$ must be rejected (Jun 22)
cot x	$\cos x/\sin x = 1/\tan x$	undefined where $\sin x = 0$
arcsin, arccos, arctan	inverse functions	ranges $[-\pi/2, \pi/2]$, $[0, \pi]$, $(-\pi/2, \pi/2)$

IDENTITIES YOU MUST RECALL

$$\sin^2\theta + \cos^2\theta \equiv 1; 1 + \tan^2\theta \equiv \sec^2\theta; 1 + \cot^2\theta \equiv \csc^2\theta.$$

WORKED EXAMPLES · EQUATIONS

Jan 22: $2 \sec^2 x - 3 \tan x = 2$ for $0 < x \leq \pi$. $2(1 + \tan^2 x) - 3 \tan x - 2 = 0 \Rightarrow \tan x(2 \tan x - 3) = 0 \Rightarrow \tan x = 0$ or $1.5 \Rightarrow x = \pi, 0.983$.

Jun 22: $3 \csc^2 2\theta - 13 \csc 2\theta - 10 = 0 \Rightarrow (3 \csc 2\theta + 2)(\csc 2\theta - 5) = 0$. Reject $-2/3$; $\sin 2\theta = 0.2 \Rightarrow \theta = 0.101, 1.47$ ($0 < \theta < \pi/2$).

Jan 22: $8 \cos \theta = 3 \csc \theta \Rightarrow 8 \sin \theta \cos \theta = 3 \Rightarrow \sin 2\theta = 3/4 \Rightarrow \theta = 24.3^\circ$.

- Never divide by a trig function that can be zero; factorise instead, and keep the solutions it gives ($\sin \theta = 0$ gave $\theta = 180^\circ$ and 360° in Oct 24).

- Adjust the range for multiple angles: for $0 < \theta < 90^\circ$, 2θ runs from 0° to 180° . Find all values of 2θ first, then halve.
- Give exactly the solutions in range: extras, or missing ones, lose the final A mark.

8 TRIGONOMETRY II: COMPOUND ANGLES AND R FORM

FORMULAE (IN THE BOOKLET, BUT LEARN THE DOUBLE ANGLES)

$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$; $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$; $\tan(A \pm B) = (\tan A \pm \tan B)/(1 \mp \tan A \tan B)$.

$\sin 2A = 2 \sin A \cos A$; $\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$; $\tan 2A = 2 \tan A/(1 - \tan^2 A)$.

$\sin 3x = 3 \sin x - 4 \sin^3 x$ (prove from $\sin(2x + x)$: Oct 20, Oct 24). $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$, $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$.

WORKED PROOFS

Jun 23: $\cos 2x/\sin x + \sin 2x/\cos x = (\cos 2x \cos x + \sin 2x \sin x)/(\sin x \cos x) = \cos x/(\sin x \cos x) = \operatorname{cosec} x$.

Oct 23: $2 \operatorname{cosec}^2 2\theta(1 - \cos 2\theta) = 2(2 \sin^2 \theta)/(4 \sin^2 \theta \cos^2 \theta) = \sec^2 \theta \equiv 1 + \tan^2 \theta$.

Start from one side and work to the other. Do not work on both sides at once or assume the result.

WORKED EXAMPLE · SOLVING WITH SIN 3X (OCT 24)

$2 \sin 3\theta = 5 \sin 2\theta$, $0 < \theta \leq 360^\circ$: $2(3s - 4s^3) = 10sc \Rightarrow s = 0$ or $6 - 8s^2 = 10c \Rightarrow 4c^2 - 5c - 1 = 0 \Rightarrow c = (5 - \sqrt{41})/8$ (the other root exceeds 1). $\theta = 100.1^\circ, 180^\circ, 259.9^\circ, 360^\circ$.

R form

METHOD (3 MARKS)

$a \cos x + b \sin x = R \cos(x - \alpha)$: $R \cos \alpha = a$, $R \sin \alpha = b \Rightarrow R = \sqrt{a^2 + b^2}$, $\tan \alpha = b/a$. Give R exact and α in radians unless told otherwise. Check the signs to place α in the stated interval.

Jun 25: $7 \cos x - 24 \sin x = 25 \cos(x + 1.287)$.

USING R FORM FOR MAX/MIN (JUN 25 Q2)

$g(x) = 5/(90 - 3f(2x))$ is smallest when the denominator is largest, i.e. when $\cos(2x + 1.287) = -1$: minimum = $5/(90 + 75) = 1/33$, at $2x + 1.287 = \pi \Rightarrow x = 0.927$.

Always ask which value of the cosine (+1 or -1) makes the whole expression a maximum or minimum, and solve for the smallest positive x in the same units as α .

- Transformations from $y = \cos \theta$ to $y = R \cos(\theta + \alpha)$ (Jan 20): a stretch parallel to the y-axis, scale factor R, then a translation by $(-\alpha, 0)$. “Translated left by -0.675 ” (a double negative) was not credited.
- Ranges such as $h(\beta) = 10 - (12 \sin \beta - 5 \cos \beta)^2$ (Jun 21): $(R \sin(\beta - \alpha))^2$ runs from 0 to 169, so $-159 \leq h \leq 10$.

9 DIFFERENTIATION

y	dy/dx	y	dy/dx
e^{kx}	ke^{kx}	$\sin kx$	$k \cos kx$
$\ln kx$	$1/x$	$\cos kx$	$-k \sin kx$
$\ln f(x)$	$f'(x)/f(x)$	$\tan kx$	$k \sec^2 kx$
a^x	$a^x \ln a$	$\sec x$	$\sec x \tan x$

y	dy/dx	y	dy/dx
$[f(x)]^n$	$n f'(x)[f(x)]^{n-1}$	$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$
uv	$u'v + uv'$	$\cot x$	$-\operatorname{cosec}^2 x$
u/v	$(vu' - uv')/v^2$	$\arcsin, \arctan$	derive via $x = \sin y$ or $x = \tan y$ if needed

WORKED EXAMPLES · STATIONARY POINTS

Jan 22: $y = (2x + 5)e^{3x}$; $dy/dx = 2e^{3x} + 3(2x + 5)e^{3x} = e^{3x}(6x + 17) = 0 \Rightarrow x = -17/6$ ($e^{3x} > 0$).

Oct 21: $y = 3 \ln(x^2 - 5) - 4x^2 + 15$; $6x/(x^2 - 5) - 8x = 0 \Rightarrow x^2 = 23/4 \Rightarrow x = \sqrt{23/4}$.

Jun 21: $y = x^2 \cos(x/2)$; $2x \cos(x/2) - \frac{1}{2}x^2 \sin(x/2) = 0 \Rightarrow \tan(x/2) = 4/x \Rightarrow x = 2 \arctan(4/x)$.

Curves given as $x = f(y)$

METHOD

Differentiate with respect to y , then $dy/dx = 1 \div (dx/dy)$. To express in x , use the curve equation and a Pythagorean identity.

Jun 21: $x = 6 \sin^2 2y \Rightarrow dx/dy = 24 \sin 2y \cos 2y$. With $\sin^2 2y = x/6$ and $\cos^2 2y = 1 - x/6$, $dx/dy = 4\sqrt{(6x - x^2)}$, so $dy/dx = 1/(4\sqrt{(6x - x^2)})$.

Jan 22: $x = ye^{2y} \Rightarrow dx/dy = e^{2y}(1 + 2y)$; since $e^{2y} = x/y$, $dy/dx = y/(x(1 + 2y))$.

Vertical tangents (Jun 23, Oct 24): $dx/dy = 0$ gives the y -values; substitute to get $x = \text{constant}$.

- Tangent/normal: gradient from dy/dx at the point; normal gradient $= -1/m$. Use $y - y_1 = m(x - x_1)$ and give the form asked, e.g. integers with “= 0”.
- Range of k for exactly two/three turning points (Oct 23, Oct 25): set $dy/dx = 0$, form a quadratic and use the discriminant ($b^2 - 4ac > 0$ for two distinct roots).
- Increasing interval (Jan 20, Jan 25): solve $f'(x) > 0$ with the factorised numerator and a positive denominator such as $(x - 3)^2$.

10 INTEGRATION

$f(x)$	$\int f(x) dx$	$f(x)$	$\int f(x) dx$
e^{ax+b}	$(1/a)e^{ax+b}$	$\sin(ax + b)$	$-(1/a) \cos(ax + b)$
$1/(ax + b)$	$(1/a) \ln ax + b $	$\cos(ax + b)$	$(1/a) \sin(ax + b)$
$(ax + b)^n$	$(ax + b)^{n+1}/(a(n + 1))$	$\sec^2 ax$	$(1/a) \tan ax$
$f'(x)/f(x)$	$\ln f(x) $	$\sin^2 x$	$\frac{1}{2}x - \frac{1}{4} \sin 2x$
$f'(x)[f(x)]^n$	$[f(x)]^{n+1}/(n + 1)$	$\cos^2 x$	$\frac{1}{2}x + \frac{1}{4} \sin 2x$

WORKED EXAMPLES

Jan 21: $\int (x^2 - 5)/(2x^3) dx = \int (1/(2x) - (5/2)x^{-3}) dx = \frac{1}{2} \ln x + 5/(4x^2) + c$. Split the fraction first; $+ c$ and simplest form are required.

Jun 22: $\int 9x/(3x^2 + k) dx = (3/2) \ln(3x^2 + k) + c$, since $d/dx(3x^2 + k) = 6x$.

Oct 20: from $\sin 3x = 3 \sin x - 4 \sin^3 x$, $\sin^3 x = \frac{1}{4}(3 \sin x - \sin 3x)$, so $\int \sin^3 x dx = -\frac{3}{4} \cos x + (1/12) \cos 3x + c$.

Oct 21: $(1 + 2 \cos 2x)^2 = 3 + 4 \cos 2x + 2 \cos 4x$. The curve first touches the x -axis at $x = \pi/3$ and again at $x = 2\pi/3$, so the area $= [3x + 2 \sin 2x + \frac{1}{2} \sin 4x]$ from 0 to $2\pi/3 = 2\pi - 3\sqrt{3}/4$.

- Exact answers: leave $\ln$, e and π in; combine logs into a single term when asked (“ $p + \ln q$ ”, “ $\alpha + \beta \ln 3$ ”).

- Areas between a curve and a line (Jun 24, Oct 24): area under the curve $\pm$ a triangle or trapezium; find limits from intersections; draw the region.
- “Use algebraic integration” means a calculator value alone scores nothing (Oct 20, Jan 22).

11 NUMERICAL METHODS

MODEL ANSWER · SHOW A ROOT LIES IN [A, B] (2)

$f(a) = \dots < 0$ and $f(b) = \dots > 0$ (values shown). There is a change of sign and f is continuous on $[a, b]$, so there is a root in $[a, b]$.

Not acceptable: “the root is continuous”, “the interval is continuous”, “there is a change of sign, therefore it is continuous”. Trig functions must be evaluated in radians.

MODEL ANSWER · VERIFY A ROOT TO 3 D.P. (2) · JAN 21, JUN 25

To show the root is 2.154 to 3 d.p., evaluate the stated function at the bounds 2.1535 and 2.1545: $f(2.1535) = 0.00084 > 0$, $f(2.1545) = -0.0026 < 0$ (Jun 21 curve). Sign change and f continuous, so $2.1535 < \alpha < 2.1545$ and $\alpha = 2.154$ to 3 d.p.

You must state the function you are using. Repeated iteration does not score these marks (Jun 25).

ITERATION (3–4) · JUN 21 Q1

$x_{n+1} = 2 \arctan(4/x_n)$, $x_1 = 2$: $x_2 = 2.214$, $x_3 = 2.130$, ..., $x_6 = 2.155$ (radians mode; degrees mode gives about 127 and loses the marks).

Write each value to the d.p. asked, using the full calculator value for the next step. Keep trailing zeros: 0.8110, not 0.811.

- Rearranging $f(x) = 0$ or $f'(x) = 0$ into the given $x = g(x)$ is a “show that”: every line of algebra must be seen, e.g. $\tan(x/2) = 4/x \Rightarrow x/2 = \arctan(4/x)$.
- Typical sources: a stationary point (Jun 21, Jan 21, Jun 24), an intersection of two curves (Oct 20), or a contextual model (Oct 22 profit, Jun 22 sprinter, Oct 24 heart rate).
- Contextual follow-up: convert the root into the answer asked, e.g. months to make a profit (Oct 22).

12 FINAL A* CHECKLIST

1. Check the mode before every trig calculation: radians unless the range is in degrees.
2. For every “show that”, write every step, start from one side, and finish with the given result exactly.
3. For every “hence”, use the previous part visibly. For “calculator technology not acceptable”, show the algebra.
4. For every inverse: in terms of x ; domain of f^{-1} = range of f . For every range: $f(x)$ or y , not x .
5. For every modulus equation: solve both cases, then check each solution against the sketch.
6. For every trig equation: factorise, do not divide; reject impossible values ($|\sec|$, $|\csc| \geq 1$); list all solutions in range and no others.
7. For every model: $t = 0$ for the initial value, $t \rightarrow \infty$ for the limit, differentiate before substituting, interpret constants as % per unit time with context.
8. For every integral: $+ c$ on indefinite answers, modulus inside $\ln$, exact form when asked.
9. For every root: two values, “change of sign”, “continuous”, “so root”; bounds ± 0.0005 for 3 d.p.
10. Leave 10 minutes: reread the accuracy demanded (3 s.f., 4 d.p., exact) for every final answer.